



Original Research Article

Quaternionic one-dimensional linear canonical transform

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ABSTRACT

The quaternion linear canonical transform has been proved a powerful tool in signal processing and optics. There are many related research work about quaternion linear canonical transform but theories about one-dimensional quaternion linear canonical transform still needs to be studied. In the present paper, we define quaternionic linear canonical transform of integrable (and square integrable) functions on R . It is also shown that it satisfies all the respective properties like inversion formula, linearity, convolution theorem, Parseval's identity and the product theorem.

1. Introduction

Linear canonical transform (LCT) [1–4] is an integral transform with four parameters a, b, c, d . It has been proved to be an influential tool in many fields. It was firstly introduced in 1970. A lot of work has been done on LCT [5–9]. The formal definition of LCT, taken from [10], is given as follows:

$$L_f^A = L_f^A[f(t)](u) = \sqrt{\frac{1}{j2\pi b}} \int_R f(t) e^{j\frac{1}{2}[\frac{a}{b}t^2 - (\frac{2}{b}tu) + \frac{d}{b}u^2]} dt, b \neq 0$$

The other transforms like Fourier transform (FT), fractional Fourier transform (FRFT), Fresnel transform (FRT) and scaling operations are being considered as the special cases of LCT [11–14]. In the light of signal processing, one can consider that any integral transform converts the signals from time domain to frequency domain. Further in signal processing, convolution of two functions [15–18] is a valuable tool in constructing a filter for denoising the noisy signals [19].

It was Hamilton who firstly gave the concept of quaternions [20]. Since then, quaternions [21–23] have been considered as a hot topic especially in signal processing. It is the generalization of the classical theory of holomorphic functions onto the multidimensional situations [21]. Recently, it has been very popular to generalize the LCT from real and complex numbers to quaternion algebra. In these generalizations, most of the LCT properties hold, rest are modified. For the representation of signals, quaternion linear canonical transform (QLCT) plays a significant role as it transfers the real or quaternionic valued 2D signals into a quaternionic-valued signals in frequency domain [24–26]. The QLCT has been shown to be a powerful tool in many fields specifically in optics and signal processing. The QLCT generalizes the quaternion Fourier transform (QFT) and the quaternion fractional Fourier transform (QFRFT) [27].

The QFT is helpful in solving heat equations under quaternion algebra. The QFRFT is used in many fields like object recognition, reconstruction and in color images [22]. Quaternion plays a significant role in the field of animation as it nicely composes the rotation and gives spherical interpolation [23]. Many fundamental properties of QLCT have been investigating but there is no

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theory about quaternionic one-dimensional LCT as far as we know, therefore it is worthwhile to study the theory of quaternionic one-dimensional LCT which can be useful for signal processing theory and applications.

In this paper, our main objectives are to propose the quaternionic one-dimensional LCT and establish some properties for the quaternionic one-dimensional LCT which are the generalizations of the related classical ones, such as inversion formula, linearity, Parseval's identity, convolution theorem and the product theorem.

The paper is organized as follows: In Section 2, we give some general definitions. The quaternionic one-dimensional LCT is provided and some properties for the quaternionic one-dimensional LCT are obtained in Section 3. In Section 4, some conclusions are drawn.

2. Preliminaries

2.1. Quaternions

The quaternion algebra, in the honor of Hamilton, is denoted by H . Quaternion is the general form of complex algebra. It is a 4D generation of complex number whose one part is considered to be real and rest of three parts are imaginary [25]. The representation of quaternion in a hypercomplex plane is as follows;

$$q = q_r + q_i i + q_j j + q_k k$$

where $q_r, q_i, q_j, q_k \in \mathbb{R}$ and $i, j, k \in \mathbb{C}$ which obey the following laws:

$$ijk = i^2 = j^2 = k^2 = -1$$

and

$$ij = -ji = k, jk = -kj = i, ki = -ik = j$$

Because of non-commutativity of quaternions, one cannot directly extend many results of complex numbers to quaternions. We can write as:

$$q = q_r + q_i i + q_j j + q_k k$$

$$q = q_r + q_s$$

where q_s is a pure 3-dimensional quaternion. The conjugate of quaternion q can simply be obtained by changing the sign of pure part q_s , i.e.

$$\bar{q} = q_r - q_s = q_r - q_i i - q_j j - q_k k$$

The multiplication of two quaternions $q = q_r + q_s$ and $p = p_r + p_s$ is given as:

$$qp = q_r p_r + q_s \cdot p_s + q_r p_s + p_r q_s + q_s \times p_s$$

where “ $\cdot$ ” and “ $\times$ ” denote respectively the dot product and the cross product.

The multiplication of quaternion with its conjugate can be expressed as:

$$q\bar{q} = q_r^2 + q_i^2 + q_j^2 + q_k^2$$

The modulus of a quaternion is:

$$|q| = \sqrt{q\bar{q}} = \sqrt{q_r^2 + q_i^2 + q_j^2 + q_k^2}$$

The inverse of $q \in H$ can be written as: $q^{-1} = \frac{\bar{q}}{|q|^2}$ with $q \neq 0$.

Here in the current section, we will collect some of the definitions and properties of linear canonical transform (LCT), quaternion Fourier transform (QFT), fractional quaternion Fourier transform (FRQFT) and QLCT.

Due to the non-commutativity of quaternions, we have the following three types of QFT, FRQFT and QLCT. The QFT is defined as [28]:

The two-sided: $F_T(r, s) = \int_{\mathbb{R}^2} e^{-iru} f(u, v) e^{-j sv} du dv$

The right-sided: $F_r(r, s) = \int_{\mathbb{R}^2} f(u, v) e^{-iru} e^{-j sv} du dv$

The left-sided: $F_l(r, s) = \int_{\mathbb{R}^2} e^{-iru} e^{-j sv} f(u, v) du dv$

For any two-dimensional function $f(x, y)$, the two sided FRQFT is defined as [29]:

$$\begin{aligned} FRF_{\beta_1 \beta_2}^{i,j}(r, s) &= FRF_{\beta_1 \beta_2}^{i,j} \{f(x, y)\} \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} K_{\beta_1}^i(x, r) f(x, y) K_{\beta_2}^j(y, s) dx dy \end{aligned}$$

where

$$K_{\beta_1}^i(x, r) = \sqrt{\frac{1 - i \cot \theta_1}{2\pi}} e^{\frac{i}{2 \sin \theta_1} [(x^2 + r^2) \cos \theta_1 - 2xr]}, \theta_1 = \beta_1 \frac{\pi}{2}$$

and

$$K_{\beta_2}^j(y, s) = \sqrt{\frac{1 - j \cot \theta_2}{2\pi}} e^{\frac{j}{2 \sin \theta_2} [(y^2 + s^2) \cos \theta_2 - 2ys]}, \theta_2 = \beta_2 \frac{\pi}{2}$$

The definition of a two-dimensional quaternionic linear canonical transform (QLCT) is taken from [17,28]. The left-sided L^l , right-sided L^r and the two-sided L^t , QLCT of a signal $f \in L^1(\mathbb{R}^2, H)$ with real parameters $A = \begin{pmatrix} a_i & b_i \\ c_i & d_i \end{pmatrix} \in \mathbb{R}^{2 \times 2}$ such that $\det(A_i) = 1$ for $i = 1, 2$ are defined as:

The left-sided: $(L^l f)(u, v) := \int_{\mathbb{R}^2} K_{A_1}^i(s, u) K_{A_2}^j(t, v) f(s, t) ds dt$

The right-sided: $(L^r f)(u, v) := \int_{\mathbb{R}^2} f(s, t) K_{A_1}^i(s, u) K_{A_2}^j(t, v) ds dt$

The two-sided: $(L^t f)(u, v) := \int_{\mathbb{R}^2} K_{A_1}^i(s, u) f(s, t) K_{A_2}^j(t, v) ds dt$
where

$$K_{A_1}^i(s, u) := \frac{1}{\sqrt{j2\pi b_1}} e^{i(\frac{a_1}{2b_1}s^2 - \frac{1}{b_1}su + \frac{d_1}{2b_1}u^2)}$$

and

$$K_{A_2}^j(t, v) := \frac{1}{\sqrt{j2\pi b_2}} e^{j(\frac{a_2}{2b_2}t^2 - \frac{1}{b_2}tv + \frac{d_2}{2b_2}v^2)}$$

It can be observed that for $b_i = 0, i = 1, 2$ the QLCT of a signal is essentially a chirp multiplication and it is of no particular use, therefore without loss of generality, we set $b_i \neq 0$. It is important to note that the QLCT reduces to special cases for different values of matrices $A_m, m = 1, 2$. For instance, $A = [a, b, c, d] = [0, 1, -1, 0]$, the QLCT reduces to the QFT times $\sqrt{\frac{-i}{2\pi}}$ and $\sqrt{\frac{-j}{2\pi}}$, with $\sqrt{-i} = e^{\frac{-i\pi}{4}}$ and $\sqrt{-j} = e^{\frac{-j\pi}{4}}$ [28]. Similarly, if we take $A_1 = [\cos \alpha, \sin \alpha, -\sin \alpha, \cos \alpha]$, $A_2 = [\cos \beta, \sin \beta, -\sin \beta, \cos \beta]$, the QLCT reduces to QFRFT times the factor $e^{\frac{-i\alpha}{2}}, e^{\frac{-j\beta}{2}}$ respectively.

3. Quaternion one-dimensional linear canonical transform

Using [29,30], we define quaternion one-dimensional linear canonical transform. For simplicity, we will write 1DQLCT instead of quaternion one-dimensional linear canonical transform.

3.1. Definition of 1DQLCT

Definition 1. 1DQLCT can be defined as

$$Q_A((x)(t))(u) = \int x(t) K_A^j(t, u) dt$$

where

$$K_A^j(t, u) := \frac{1}{\sqrt{j2\pi b}} e^{j(\frac{a}{2b}u^2 - \frac{1}{b}tu + \frac{d}{2b}t^2)}$$

This is the definition of 1DQLCT which reduces to quaternion one-dimensional Fourier transform for $A = [a, b, c, d] = [0, 1, -1, 0]$ and for $A = [a, b, c, d] = [\cos \alpha, \sin \alpha, -\sin \alpha, \cos \alpha]$, the proposed Definition 1 reduces to quaternionic one-dimensional fractional Fourier transform times the factor $e^{\frac{-i\alpha}{2}}$ [29,30]. In classical definition, the signal $f(x) \in L(\mathbb{R})$ while in the proposed definition the signal $f(x) \in L(\mathbb{R}, H)$.

There are many papers about two-dimensional QLCT, for example [26], but the proposed paper is about QLCT in one-dimension.

Definition 2. For a function $x = x_1 + ix_2 \in L^1(\mathbb{R}, H)$ in the quaternionic domain, we give another definition of QLCT Q_A as:

$$Q_A(x) = Q_A(x_1) + iQ_A(x_2)$$

This definition is consistent with the LCT on $L^1(\mathbb{R}, C)$. By the definition of QLCT and properties of LCT on $L^1(\mathbb{R}, H)$, we can verify that $Q_A(Q_B(x)) = Q_{AB}(x)$ for all $f \in L^1(\mathbb{R}, H)$ and $Q_B(x) \in L^1(\mathbb{R}, H)$, so $Q_A^{-1} = Q_{A^{-1}}$ for all $x \in L^1(\mathbb{R}, H)$.

3.2. Properties of 1DQLCT

Theorem 1 (H-Linearity). The QLCT Q_A is H -linear on $L^1(\mathbb{R}, H)$.

Proof. Let $x = x_1 + ix_2, y = y_1 + iy_2 \in L^1(\mathbb{R}, H)$, then by using the linearity of Q_A on $L^1(\mathbb{R}, C)$, we get

$$\begin{aligned} Q_A(x + y) &= Q_A[(x_1 + ix_2) + (y_1 + iy_2)] \\ &= Q_A[(x_1 + y_1) + i(x_2 + y_2)] \\ &= Q_A(x_1) + Q_A(y_1) + i[Q_A(x_2) + Q_A(y_2)] \\ &= (Q_A(x_1) + iQ_A(x_2)) + (Q_A(y_1) + iQ_A(y_2)) \\ &= Q_A(x) + Q_A(y) \end{aligned} \tag{1}$$

To prove H-linearity, let $p = p_1 + ip_2 \in H$, $x = x_1 + ix_2 \in L^1(R, H)$ be arbitrary, then we have

$$\begin{aligned} Q_A(ix) &= Q_A(i(x_1 + ix_2)) \\ &= Q_A(ix_1 - x_2) \\ &= iQ_A(x_1) - Q_A(x_2) \\ &= i(Q_A(x_1) + iQ_A(x_2)) \\ &= iQ_A(x) \end{aligned} \quad (2)$$

Therefore, we obtain

$$\begin{aligned} Q_A(px) &= Q_A(p_1x) + Q_A(ip_2x) \\ &= p_1Q_A(x) + ip_2Q_A(x) \\ &= (p_1 + ip_2)Q_A(x) \\ &= pQ_A(x) \end{aligned} \quad (3)$$

This completes the proof.

Theorem 2 (Parseval's Identity). For $x, y \in L^1(R, H) \cap L^2(R, H)$ with $Q_A(x) \in L^1(R, H)$, $\langle x, y \rangle = \langle Q_A(x), Q_A(y) \rangle$.

Proof. For $x, y \in L^1(R, H) \cap L^2(R, H)$ with $Q_A(x) \in L^1(R, H)$, we use inversion formula

$$\begin{aligned} \langle x, y \rangle &= \int x(t)[y(t)]^c dt \\ &= \iint Q_A(x)(u)K_{A^{-1}}^j(u, t)du[y(t)]^c dt \\ &= \iint Q_A(x)(u)K_{A^{-1}}^j(u, t)[y(t)]^c dt du \end{aligned} \quad (4)$$

According to $K_{A^{-1}}^j(u, t) = [K_A^j(t, u)]^c$, we have

$$\begin{aligned} \langle x, y \rangle &= \int Q_A(x)(u) \int [y(t)K_A^j(t, u)]^c dt du \\ &= \int Q_A(x)(u) \left[\int y(t)K_A^j(t, u) du \right]^c dt \\ &= \int Q_A(x)(u)[Q_A(y)]^c dt \\ &= \langle Q_A(x), Q_A(y) \rangle \end{aligned} \quad (5)$$

This completes the proof.

Lemma 1. For $x \in L^p(R, C)$, $p = 1, 2$; we get $\overline{Q_A(x)} = Q_{A^{-1}}(\bar{x})$, $\forall t \in R$

Proof.

$$\begin{aligned} \overline{Q_A(x)(u)} &= \overline{\int x(t)K_A^j(t, u)dt} \\ &= \int \overline{x(t)} \frac{1}{\sqrt{j2\pi b}} e^{j(\frac{a}{2b}u^2 - \frac{ut}{b} + \frac{d}{2b}t^2)} dt \\ &= \int \overline{x(t)} \frac{1}{\sqrt{-j2\pi b}} e^{-j(\frac{a}{2b}u^2 - \frac{ut}{b} + \frac{d}{2b}t^2)} dt \\ &= \int \overline{x(t)} K_{A^{-1}}^j(u, t) dt \\ &= Q_{A^{-1}}(\bar{x}) \end{aligned} \quad (6)$$

This completes the proof.

Let $G = \{x_1 + ix_2 : x_1, x_2 \in C\}$, where i is an imaginary number other than the imaginary complex number and it satisfies the following properties $i^2 = -1$, $is = si$, for all $s \in R$, $ij = -ji$, $j \in C$ [31]. All elements of G are quaternions and satisfy the addition, multiplication, conjugation and absolute value properties as follows:

$$\begin{aligned} (x_1 + ix_2) + (y_1 + iy_2) &= (x_1 + y_1) + i(x_2 + y_2), \\ (x_1 + ix_2)(y_1 + iy_2) &= (x_1y_1 - \bar{x}_2y_2) + i(x_2y_1 + \bar{x}_1y_2), \\ (x_1 + ix_2)^c &= \bar{x}_1 - ix_2, \end{aligned}$$

$$|x_1 + ix_2| = \sqrt{|x_1|^2 + |x_2|^2},$$

where $\bar{x}_k$ is complex conjugate and $|x_k|$ is the modulus of complex number $x_k, k = 1, 2$. The well known properties of absolute value and conjugate are given as:

$$(x^c)^c = x$$

$$(x + y)^c = x^c + y^c$$

$$(xy)^c = x^c y^c$$

$$|x|^2 = xx^c = |x_1|^2 + |x_2|^2$$

$$|xy| = |x||y|$$

3.3. Convolution and product theorems for 1DQLCT

In this subsection, we will obtain the convolution and product theorems for 1DQLCT.

Definition 3. For $x \in L^2(R, H)$ and $y \in L^1(R, H)$, we define

$$(x * y)(t) = (Q_{A^{-2}\bar{x}_1} * y_1 + x_2 * y_2) + i(Q_{A^{-2}\bar{x}_1} * y_2 - x_2 * y_1)$$

where $*$ is the proposed definition of convolution.

Lemma 2. For $x \in L^2(R, H)$ and $y \in L^1(R, H)$, we have

$$Q_A Q_B = Q_{AB}$$

$$Q_A(x * y)(u) = \overline{Q_A(x)(u)} Q_A(y)(u) \sqrt{2j\pi be}^{-\frac{jdu^2}{2b}} \forall u \in R$$

$$Q_A(x(t)y(t)e^{\frac{jdt^2}{2b}}) = (Q_A(x) \otimes Q_A(y))(u) \forall u \in R$$

where $Q_A(x) \otimes Q_A(y) = Q_A(xy)$

Theorem 3 (Convolution Theorem). For $x \in L^2(R, H)$ and $y \in L^1(R, H)$, we have $Q_A(x * y)(t) = \overline{Q_A(x)} Q_A(y) \sqrt{j2\pi be}^{-\frac{jdt^2}{(2b)}}$.

Proof. From Definition 3 and Lemma 2, we have

$$\begin{aligned} Q_A(x * y)(t) &= Q_A(Q_{A^{-2}\bar{x}_1} * y_1 + x_2 * y_2)(t) + iQ_A(Q_{A^{-2}\bar{x}_1} * y_2 - x_2 * y_1) \\ &= Q_A Q_{A^{-2}\bar{x}_1}(t) Q_A y_1(t) \sqrt{j2\pi be}^{-\frac{jdt^2}{(2b)}} + iQ_A Q_{A^{-2}\bar{x}_1}(t) Q_A y_2(t) \sqrt{j2\pi be}^{-\frac{jdt^2}{(2b)}} \\ &\quad - iQ_A x_2(t) Q_A y_1(t) \sqrt{j2\pi be}^{-\frac{jdt^2}{(2b)}} + Q_A x_2(t) Q_A y_2(t) \sqrt{j2\pi be}^{-\frac{jdt^2}{(2b)}} \\ &= \{[Q_{A^{-1}\bar{x}_1} Q_A y_1 + Q_A x_2 Q_A y_2](t) + i[Q_{A^{-1}\bar{x}_1} Q_A y_2 - Q_A x_2 Q_A y_1](t)\} \sqrt{j2\pi be}^{-\frac{jdt^2}{(2b)}} \\ &= \{[\overline{Q_A x_1} Q_A y_1 + Q_A x_2 Q_A y_2](t) + i[\overline{Q_A x_1} Q_A y_2 - Q_A x_2 Q_A y_1](t)\} \sqrt{j2\pi be}^{-\frac{jdt^2}{(2b)}} \\ &= \overline{Q_A(x)(t)} Q_A(y)(t) \sqrt{j2\pi be}^{-\frac{jdt^2}{(2b)}} \end{aligned} \tag{7}$$

Hence, the result is proved.

Definition 4. For $x = x_1 + ix_2 \in L^2(R, H)$, $y = y_1 + iy_2 \in L^1(R, H)$, we can give the definition as :

$$x \theta y = \overline{Q_{A^{-2}\bar{x}_1}} \otimes y_1 + x_2 \otimes y_2 + i[\overline{Q_{A^{-2}\bar{x}_1}} \otimes y_2 - x_2 \otimes y_1]$$

where $Q_A(\bar{x}_1) \otimes Q_A(y_1) = Q_A(\bar{x}_1 y_1)$

Theorem 4 (Product Theorem). For $x, y \in L^2(R, H)$, $Q_A(\bar{x}y) = Q_A(x) \theta Q_A(y)$.

Proof. According to Definition 4 and Lemma 2, we obtain

$$\begin{aligned}
 Q_A(\bar{x}y) &= Q_A((\bar{x}_1 - ix_2)(y_1 + y_2)) \\
 &= Q_A(\bar{x}_1 y_1) + Q_A(x_2 y_2) + i[Q_A(\bar{x}_1 y_2) - Q_A(x_2 y_1)] \\
 &= Q_A(\bar{x}_1) \otimes Q_A(y_1) + Q_A(x_2) \otimes Q_A(y_2) \\
 &\quad + i[Q_A(\bar{x}_1) \otimes Q_A(y_2) - Q_A(x_2) \otimes Q_A(y_1)] \\
 &= \overline{Q_{A^{-1}}(x_1)} \otimes Q_A(y_1) + Q_A(x_2) \otimes Q_A(y_2) \\
 &\quad + i[\overline{Q_{A^{-1}}(x_1)} \otimes Q_A(y_2) - Q_A(x_2) \otimes Q_A(y_1)] \\
 &= \overline{Q_{A^{-2}}Q_A(x_1)} \otimes Q_A(y_1) + Q_A(x_2) \otimes Q_A(y_2) \\
 &\quad + i[\overline{Q_{A^{-2}}Q_A(x_1)} \otimes Q_A(y_2) - Q_A(x_2) \otimes Q_A(y_1)] \\
 &= Q_A(x) \otimes Q_A(y)
 \end{aligned} \tag{8}$$

The result is proved.

4. Conclusions

In this paper, we have proposed the definition of quaternion one-dimensional LCT. The related product theorem, Parseval's identity and the convolution theorem are also derived. This work is the generalization of LCT of complex valued functions to quaternionic domain. Our future work about sampling of quaternionic one-dimensional LCT is in progress.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Y.N. Sun, B.Z. Li, R. Tao, Research progress on discretization of linear canonical transform, *Opto-Electron. Eng.* 45 (6) (2018) 170738.
- [2] Y.N. Sun, B.Z. Li, Digital computation of linear canonical transform for local spectra with flexible resolution ability, *Sci. China Inf. Sci.* 62 (2019) 49301, <http://dx.doi.org/10.1007/s11432-018-9585-1>.
- [3] M. Moshinsky, Linear canonical transformations and their unitary representations, *J. Math. Phys.* 12 (8) (1971) 1772–1780.
- [4] H. John, A.K.M. J., M.O. Haldun, T.S.e. John, *Linear Canonical Transforms: Theory and Applications*, Springer, 2016.
- [5] B.Z. Li, S. Yu Pu, Image watermarking in the linear canonical transform domain, *Math. Probl. Eng.* 2014 (645059) (2014) 1–9.
- [6] Q. Feng, B.Z. Li, Convolution and correlation theorems for the two dimensional linear canonical transform and its applications, *IET Signal Process.* 10 (2) (2016) 125–132.
- [7] Y.N. Zhang, B.Z. Li, Phi-linear canonical analytic signals, *Signal Process.* 143 (2018) 181–190.
- [8] Y.N. Sun, B.Z. Li, Segmented fast linear canonical transform, *J. Opt. Soc. Am. A-Opt. Imag. Vision* 35 (8) (2018) 1346–1355.
- [9] Z.C. Zhang, New convolution structure for linear canonical transform and its application in filter design, *Optik* 127 (1) (2016) 5259–5263.
- [10] D. Wei, Q. Ran, Y. Li, New convolution theorem for the linear canonical transform and its translation invariance property, *Optik* 123 (9) (2012) 1478–1481.
- [11] B.Z. Li, R. Tao, Y. Wang, New sampling formulae related to linear canonical transform, *Signal Process.* 87 (5) (2007) 983–990.
- [12] A. Stern, Sampling of linear canonical transformed signals, *Signal Process.* 86 (2006) 1421–1425.
- [13] L. Zhao, J.T. Sheridan, J.J. Healy, Unitary algorithm for nonseparable linear canonical transforms applied to iterative phase retrieval, *IEEE Signal Process. Lett.* 24 (6) (2017) 814–817, <http://dx.doi.org/10.1109/LSP.2017.2684829>.
- [14] D. wei, W. Yang, Y.M. Li, Lattices sampling and sampling rate conversion of multidimensional bandlimited signals in the linear canonical transform domain, *J. Franklin Inst. B* (ISSN: 0016-0032) 13 (8) (2019).
- [15] D. Urynbassarova, B.Z. Li, R. Tao, Convolution and correlation theorems for wigner-ville distribution associated with offset linear canonical transform, *Optik* 157 (2018) 455–466.
- [16] D. Wei, Y.-M. Li, Generalized wavelet transform based on the convolution operator in the linear canonical transform domain, *Optik* 125 (2014) 4491–4496.
- [17] M. Bahri, R. Ashino, R. Vaillancourt, Convolution theorems for quaternion fourier transform: Properties and applications, in: *Hindawi Publishing Corporation Abstract and Applied Analysis*, 2013, 2013, p. 10.
- [18] V.D. Sharma, P.B. Deshmukh, Convolution structure of fractional quaternion fourier transform, *IJESRT* (2016) 176–182.
- [19] Z. Zhang, Linear canonical wigner distribution based noisy LFM signals detection through the output SNR improvement analysis, *IEEE Trans. Signal Process.* 67 (2019) 5527–5542.
- [20] W. Hamilton, *Elements of Quaternions*, 1864.
- [21] A.S. Hardy, A.S. Christie, *Elements of quaternions*, *Science* 2 (75) (1969) 564.
- [22] A.S. Gudadhe, P.P. Thakare, Introduction to quaternion linear canonical transform, *J. Sci. Arts* 14 (1) (2014) 45–52.
- [23] M. Bahri, Quaternion linear canonical transform application, *Glob. J. Pure Appl. Math.* 11 (1) (2015) 19–24.
- [24] Y.N. Zhang, B.Z. Li, Novel uncertainty principles for two sided quaternion linear canonical transform, *Adv. Appl. Clifford Algebr.* 28, 15.
- [25] M. Bahri, E.S.M. Hitzer, A. Hayashi, R. Ashino, An uncertainty principle for quaternion Fourier transform, *Comput. Math. Appl.* 56 (9) (2008) 2398–2410.
- [26] M. Bahri, R. Ashino, Two-dimensional quaternion linear canonical transform: Properties, convolution, correlation, and uncertainty principle, *Hindawi J. Math.* 2019 (2019) 1062979, <http://dx.doi.org/10.1155/2019/1062979>, 13.
- [27] W.B. Gao, B.Z. Li, Quaternion windowed linear canonical transform of two-dimensional signals, *Adv. Appl. Clifford Algebr.* 30 (16) (2020).
- [28] H. Huo, A new convolution theorem associated with the linear canonical transform, *Signal Image Video Process.* 13 (2019) 127–133.
- [29] R. Roopkumar, Quaternionic one-dimensional fractional fourier transform, *Optik* 127 (2016) 11657–11661.
- [30] Mawardi Bahri, et al., On one-dimensional quaternion fourier transform, *Phys.: Conf. Ser.* 1341 (2019) 062004.
- [31] K.I. Kou, X.X. Hu, Quaternion fourier and linear canonical inversion theorems.